National University of Computer and Emerging Sciences

Lahore Campus

Fundamentals of Databases (CS2011)

Date: Nov 3rd 2025

Course Instructor(s)

Shazia Haque

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 2

Roll No _____ Section _____ Student Signature _____

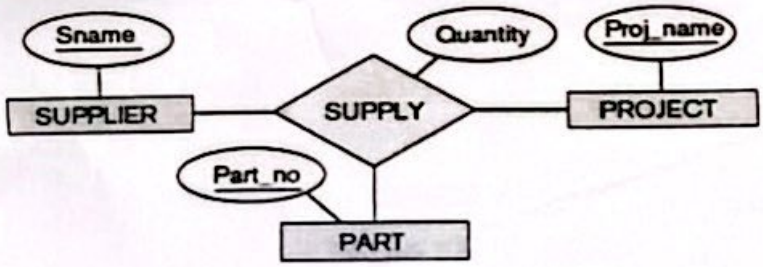
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Please answer all parts of a question together

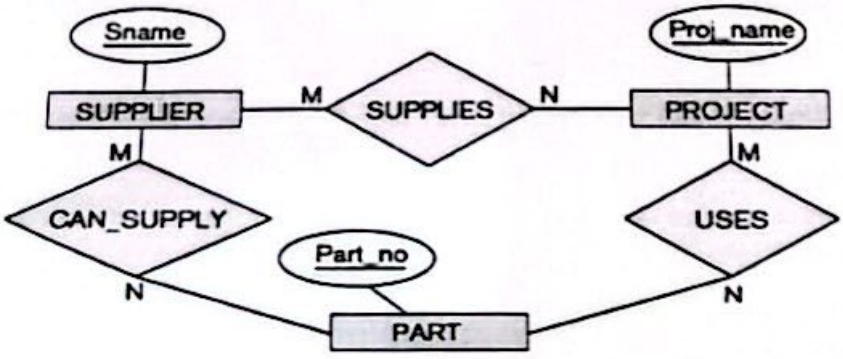
CLO # 3, PLO2, C4: Analyze system requirements and come up with high level conceptual data model

Q1: [20 marks]

a) After analyzing a system's requirements I came up with the following ER diagram



The modeling tool I used did not allow me to have a relationship of degree greater than 2. I redesigned the model as follows



Please **analyze and verify** the above model and state whether it has any issues. If it does have any issues, then please give a better solution.

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b) **Analyze** the following system requirements for a ONLINE_AUCTION system and **lay out** an EER diagram

The online site has members, each of whom is identified by a unique member number and is described by an email address, name, password, home address and phone number.

A member may be a buyer or a seller. A buyer has a shipping address recorded in the database. A seller has a bank account number and routing number recorded in the database.

Items are placed by a seller for sale and are identified by a unique item number assigned by the system. Items are also described by an item title, description, starting bid price, the start date of the auction and the end date of the auction.

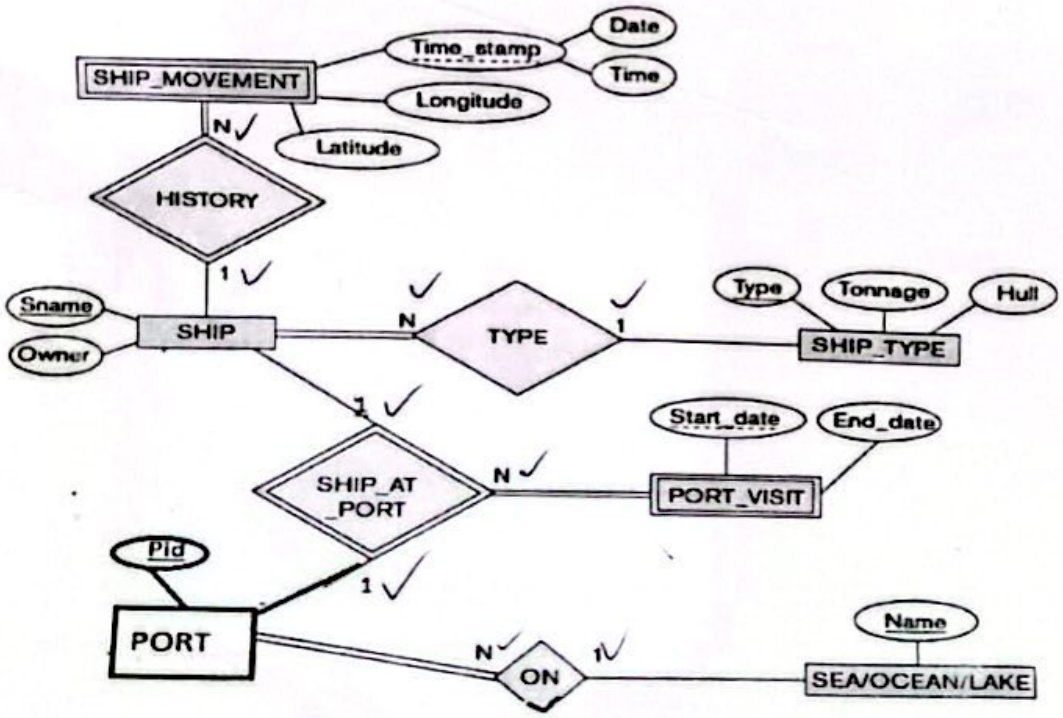
Buyers make bids for items they are interested in. Bid price and time of bid is recorded. The bidder at the end of the auction with the highest bid price is declared the winner and a transaction between buyer and seller may then proceed.

The buyer and seller may record feedback regarding their completed transactions. Feedback contains a rating of the other party participating in the transaction (1-10) and a comment.

CLO # 4, PLO1, C3: Apply the rules that are needed to convert an ER/EER model into a corresponding relational data model

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Q2: [10marks] Derive the relational model against the ER schema given below



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